

# MATH 1342 – Elementary Statistics – Week 1: Introduction and the Structure of Data

Aug 25 – 27 | Introduction: What Is Statistics For? • 1.1 The Structure of Data | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                        | Instructional Materials                                                                                                                     | Learning Activities                                                                                                                                              | Assessments                                                                  | Tools                                                             |
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| <p><b>1.</b> Describe statistics as the process of collecting, describing, and analyzing data, and judge a striking result by how likely it is to occur by chance alone. (CLO 1, CLO 4)</p> <p><b>2.</b> Distinguish a sample from a population and explain how a non-representative sample biases an estimate. (CLO 1)</p> <p><b>6.</b> Compute and compare the mean and median of a small dataset and explain how an outlier affects each. (CLO 2)</p> | <p>Introduction Slides: What Is Statistics For? (Day 1)</p>                                                                                 | <p>Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age</p> |                                                                              | <p>Blackboard<br/>Zoom (live class)</p>                           |
| <p><b>3.</b> Identify the cases and variables in a dataset presented as a table. (CLO 2)</p> <p><b>4.</b> Classify a variable as categorical or quantitative. (CLO 2)</p> <p><b>5.</b> Identify the explanatory and response variables in a research question. (CLO 1)</p>                                                                                                                                                                               | <p>e-Text: 1.1 The Structure of Data</p> <p>Interactive Lecture: 1.1 The Structure of Data</p> <p>Slides: 1.1 The Structure of Data</p>     | <p>Participation: Variables – identify cases, variables and variable types</p>                                                                                   | <p>Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM)</p> | <p>WileyPLUS<br/>StatKey<br/>Blackboard<br/>Zoom (live class)</p> |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                  | <p>START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies &amp; Student Support Services</p> | <p>Participation: Syllabus – course tour and expectations</p>                                                                                                    |                                                                              | <p>Content Area in Blackboard<br/>Zoom (live class)</p>           |

**CLO 1** Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 2** Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 4** Explain the role of probability in statistics.

# MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Sep 1 – 3 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                | Instructional Materials                                                                                                                                                                                                 | Learning Activities                                                                                                                                                                                                        | Assessments                                                                                 | Tools                                                                                     |
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| <p>1. Distinguish a population from a sample and describe statistical inference. (CLO 1)</p> <p>2. Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CLO 1)</p> <p>3. Recognize bias from question wording, question context, and inaccurate responses. (CLO 1)</p> | <p>e-Text: 1.2 Sampling from a Population</p> <p>Interactive Lecture: 1.2 Sampling from a Population</p> <p>Slides: 1.2 Sampling from a Population (includes sampling video)</p>                                        | <p>Participation: Variables in Statistics (Kahoot!) – Tue: review of Week 1</p> <p>Participation: Sampling – spot the bias in sample scenarios</p> <p>Participation: Sampling Methods (Kahoot!) – Thu: sampling review</p> | <p>Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)</p>            | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Kahoot!</p> <p>Zoom (live class)</p> |
| <p>4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CLO 1)</p> <p>5. Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CLO 1)</p>                                                        | <p>e-Text: 1.3 Experiments and Observational Studies</p> <p>Interactive Lecture: 1.3 Experiments and Observational Studies</p> <p>Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)</p> | <p>Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment</p>                                                                                         | <p>Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>                |
| <p><i>Items that cover objectives from more than one row above (objectives in parentheses)</i></p>                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                         |                                                                                                                                                                                                                            | <p>Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives)</p>  | <p>Assignment Tool in Blackboard</p>                                                      |

**CLO 1** Explain the use of data collection and statistics as tools to reach reasonable conclusions.

**MATH 1342 – Elementary Statistics – Week 3: Describing Categorical and Quantitative Variables**

**Sep 8 – 10** | 2.1 Categorical Variables • 2.2 One Quantitative Variable: Shape and Center | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Instructional Materials                                                                                                                             | Learning Activities                                                                                                                                                    | Assessments                                                                      | Tools                                                                      |
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| <p><b>1.</b> Construct frequency and relative frequency tables, compute a sample proportion <math>\hat{p}</math>, and distinguish the parameter <math>p</math> from the statistic <math>\hat{p}</math>. (CLO 2)</p> <p><b>2.</b> Compute and interpret overall and conditional proportions, and a difference in proportions, from a two-way table, choosing the correct denominator. (CLO 2)</p> <p><b>3.</b> Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CLO 2)</p> | <p>e-Text: 2.1 Categorical Variables</p> <p>Interactive Lecture: 2.1 Categorical Variables</p> <p>Slides: 2.1 Categorical Variables</p>             | <p>Participation: Frequency and Relative Frequency</p> <p>Participation: Finding Proportions from Two-Way Tables</p>                                                   | <p>Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM)</p>     | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p> |
| <p><b>4.</b> Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CLO 2)</p> <p><b>5.</b> Compute the mean and median of a dataset and explain how skewness and outliers affect each (resistance). (CLO 2)</p> <p><b>6.</b> Interpret the standard deviation as the typical distance from the mean and apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CLO 2)</p>                                             | <p>e-Text: 2.2 One Quantitative Variable</p> <p>Interactive Lecture: 2.2 One Quantitative Variable</p> <p>Slides: 2.2 One Quantitative Variable</p> | <p>Participation: Quantitative Variables – shape, mean vs. median</p> <p>Estimation activity: estimate the mean and SD of a dotplot without calculating (95% rule)</p> | <p>Online Homework: 2.2 One Quantitative Variable (due Sun Sep 13, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p> |

**CLO 2** Recognize, examine, and interpret the basic principles of describing and presenting data.

**MATH 1342 – Elementary Statistics – Week 4: Percentiles, Boxplots and Two-Variable Relationships**

**Sep 15 – 17** | 2.3 One Quantitative Variable: Percentiles • 2.4 Two Variable Relationships | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                      | Instructional Materials                                                                                                                                                                                      | Learning Activities                                                                                                                                                                                                                                  | Assessments                                                                                         | Tools                                                                                     |
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| <p><b>1.</b> Compute the sample standard deviation of a small dataset by hand. (CLO 2)</p> <p><b>2.</b> Compute and interpret z-scores to compare values measured on different scales or in different groups. (CLO 2)</p> <p><b>3.</b> Find the five-number summary, range, and IQR; apply the 1.5×IQR rule to identify outliers; and construct or interpret a boxplot. (CLO 2)</p>                                                                    | <p>e-Text: 2.3 One Quantitative Variable: Using Percentiles</p> <p>Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles</p> <p>Slides: 2.3 One Quantitative Variable: Using Percentiles</p> | <p>Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2</p> <p>Participation: Using Percentiles – Your Turn: comparing heights with z-scores</p> <p>Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule</p> | <p>Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Kahoot!</p> <p>Zoom (live class)</p> |
| <p><b>4.</b> Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CLO 2)</p> <p><b>5.</b> Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CLO 7)</p> <p><b>6.</b> Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CLO 7)</p> | <p>e-Text: 2.4 Two Variable Relationships</p> <p>Slides: 2.4 Two Variable Relationships</p>                                                                                                                  | <p>Quick Self-Quizzes: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier</p>                                                                                            | <p>Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM)</p>                   | <p>WileyPLUS</p> <p>Blackboard</p> <p>Zoom (live class)</p>                               |
| <p><i>Items that cover objectives from more than one row above (objectives in parentheses)</i></p>                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                      | <p>Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.4 (all objectives)</p>                | <p>Assignment Tool in Blackboard</p>                                                      |

**CLO 2** Recognize, examine, and interpret the basic principles of describing and presenting data. **CLO 7** Solve linear regression and correlation problems.

# MATH 1342 – Elementary Statistics – Week 5: Linear Regression

Sep 22 – 24 | 2.5 Two Quantitative Variables: Linear Regression • Catch-Up Day (Thu Sep 24) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Instructional Materials                                                                                                                                                                                                                                                                                                         | Learning Activities                                                                                                                                                                                                                                                                                        | Assessments                                                                                          | Tools                                                                              |
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| <p>1. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CLO 7)</p> <p>2. Use a regression equation to predict a response value. (CLO 7)</p> <p>3. Compute and interpret a residual, and locate positive and negative residuals on a scatterplot. (CLO 7)</p> <p>4. Use technology to find the least-squares regression line and explain what “least squares” means. (CLO 7)</p> <p>5. Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CLO 7)</p> | <p>e-Text: 2.5 Two Quantitative Variables: Linear Regression</p> <p>Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1)</p> <p>Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2)</p> <p>Slides: 2.5 Two Quantitative Variables: Linear Regression</p> | <p>Participation: Two Variable Relationships – Tue: correlation review, Your Turn: spotting residuals</p> <p>Participation: Regression Line – Thu: baseball game length and used-car price problems</p> <p>Example 2 (commute distance and time): find the regression line with technology and predict</p> | <p>Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>         |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                            | <p>Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice)</p>           | <p>Assessment Tool in Blackboard</p> <p>Respondus LockDown Browser and Monitor</p> |

**CLO 7** Solve linear regression and correlation problems.

# MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 29 – Oct 1 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Instructional Materials                                                                                                                    | Learning Activities                                                                                                                                                                                                              | Assessments                                                                                                                                               | Tools                                                                                                                   |
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| <p><b>1.</b> List the sample space and count outcomes to compute probabilities for equally likely outcomes, and apply the complement rule. (CLO 3)</p> <p><b>2.</b> Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CLO 3)</p>                                                                                                                                                                                                                       | <p>e-Text: P.1 Probability Rules</p> <p>Interactive Lecture: P.1 Probability Rules</p> <p>Slides: P.1 Probability Rules</p>                | <p>Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement</p>                                                                                                                 | <p>Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)</p>                                                                                   | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>                                              |
| <p><b>3.</b> Classify a numerical summary as a parameter or a statistic and use the correct notation (<math>\mu</math>, <math>\sigma</math>, <math>p</math>, <math>\rho</math> vs. <math>\bar{x}</math>, <math>s</math>, <math>\hat{p}</math>, <math>r</math>). (CLO 5)</p> <p><b>4.</b> Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CLO 5)</p> <p><b>5.</b> Explain the standard error as the variability of a statistic and predict how sample size affects it. (CLO 5)</p> | <p>e-Text: 3.1 Sampling Distributions</p> <p>Interactive Lecture: 3.1 Sampling Distributions</p> <p>Slides: 3.1 Sampling Distributions</p> | <p>Participation: Sampling Distributions – M&amp;M activity: each student draws 10 M&amp;Ms, computes <math>\hat{p}</math> (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents</p> | <p>Online Homework: 3.1 Sampling Distributions (due Sun Oct 4, 11:59 PM)</p>                                                                              | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>                                              |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                            |                                                                                                                                                                                                                                  | <p>Take-Home Quiz 3 (due Tue Sep 29, 11:59 PM) – covers 2.5</p> <p>Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.</p> | <p>Assignment Tool in Blackboard</p> <p>Assessment Tool in Blackboard</p> <p>Respondus LockDown Browser and Monitor</p> |

**CLO 3** Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables.

# MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 6 – 8 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                           | Instructional Materials                                                                                                                                                                                                                                                                                                                                    | Learning Activities                                                                                                                        | Assessments                                                                                                | Tools                                                                              |
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| <p><b>1.</b> Construct a 95% confidence interval for a mean or proportion as statistic <math>\pm 2 \cdot \text{SE}</math>, given the statistic and its standard error. (CLO 6)</p> <p><b>2.</b> Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CLO 4, CLO 6)</p>                                                                             | <p>e-Text: 3.2 Understanding and Interpreting Confidence Intervals</p> <p>Interactive Lecture: 3.2 Understanding and Interpreting Confidence Intervals</p> <p>Slides: 3.2 Understanding and Interpreting Confidence Intervals</p>                                                                                                                          | <p>Participation: Interpreting Confidence Intervals – which of 20 simulated CIs capture <math>\mu</math>; fix three misinterpretations</p> | <p>Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>         |
| <p><b>3.</b> Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CLO 5, CLO 6)</p> <p><b>4.</b> Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CLO 5, CLO 6)</p> <p><b>5.</b> Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6)</p> | <p>e-Text: 3.3 Constructing Bootstrap Confidence Intervals</p> <p>Interactive Lecture: 3.3 Constructing Bootstrap Confidence Intervals</p> <p>Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator)</p> <p>StatKey (<a href="https://lock5stat.com/statkey">lock5stat.com/statkey</a>) – Bootstrap Confidence Intervals</p> | <p>Participation: Bootstrap a Proportion – Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI</p> | <p>Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)</p>         | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>         |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                            | <p>Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)</p>                              | <p>Assessment Tool in Blackboard</p> <p>Respondus LockDown Browser and Monitor</p> |

**CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.

**MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests**

**Oct 13 – 15** | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Instructional Materials                                                                                                                                                                                                  | Learning Activities                                                                                                                                                                                                                   | Assessments                                                                                             | Tools                                                             |
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| <p><b>1.</b> Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CLO 3, CLO 6)</p> <p><b>2.</b> Predict how changing the confidence level or the sample size affects the width of an interval. (CLO 6)</p> <p><b>3.</b> Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CLO 6)</p> | <p>e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles</p> <p>Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles</p> <p>Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles</p> | <p>Participation: Percentile Confidence Intervals – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods</p>                                                                                             | <p>Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM)</p> | <p>WileyPLUS<br/>StatKey<br/>Blackboard</p>                       |
| <p><b>4.</b> Identify the parameter of interest and write null and alternative hypotheses from a research question. (CLO 8)</p> <p><b>5.</b> Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming <math>H_0</math> and asking whether the data are surprising. (CLO 8)</p>                                                                                                                                                                | <p>e-Text: 4.1 Introducing Hypothesis Tests</p> <p>Interactive Lecture: 4.1 Introducing Hypothesis Tests</p> <p>Slides: 4.1 Introducing Hypothesis Tests</p>                                                             | <p>Participation: Writing Hypotheses – Zener-card ESP practice (26 of 50): define the parameter, state <math>H_0</math> and <math>H_a</math>, choose the tail<br/>Think about it: is 15 of 16 correct (dolphin study) convincing?</p> | <p>Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM)</p>                     | <p>WileyPLUS<br/>StatKey<br/>Blackboard<br/>Zoom (live class)</p> |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                       | <p>Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers P.1, 3.1–3.3</p>                                | <p>Assignment Tool in Blackboard</p>                              |

**CLO 3** Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance**

**Oct 20 – 22** | 4.2 Measuring Evidence with p-values • 4.3 Determining Statistical Significance | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Instructional Materials                                                                                                                                                                                                                                                                                                                                                                                                                             | Learning Activities                                                                                                                                                                                                                                                                                                                                                                 | Assessments                                                                                                                                                                                                                                                  | Tools                                                                                                           |
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| <p><b>1.</b> Describe how a randomization distribution is generated assuming <math>H_0</math> is true, and identify its center as the null value. (CLO 5, CLO 8)</p> <p><b>2.</b> Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5)</p> <p><b>3.</b> Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from <math>H_a</math>. (CLO 3, CLO 8)</p> <p><b>4.</b> Interpret a p-value in context and relate its size to the strength of evidence against <math>H_0</math>. (CLO 4, CLO 8)</p> <p><b>5.</b> Compare the p-value to <math>\alpha</math>, decide whether to reject <math>H_0</math>, write a two-step conclusion in context, and avoid saying “accept <math>H_0</math>”. (CLO 8)</p> | <p>e-Text: 4.2 Measuring Evidence with P-Values</p> <p>Interactive Lecture: 4.2 Measuring Evidence with P-Values</p> <p>Slides: 4.2 Measuring Evidence with P-Values</p> <p>e-Text: 4.3 Determining Statistical Significance</p> <p>Interactive Lecture: 4.3 Determining Statistical Significance</p> <p>Slides: 4.3 Determining Statistical Significance (includes “What is a p-value?” video)</p> <p>StatKey – Randomization Hypothesis Tests</p> | <p>Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed</p> <p>Participation: P-values and Conclusions – resveratrol examples (<math>p = 0.04, 0.12, 0.0003</math>): strength of evidence, decision at <math>\alpha</math>, conclusion in context</p> | <p>Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM)</p> <p>Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM)</p> <p>Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 3.4, 4.1–4.3</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p> <p>Assignment Tool in Blackboard</p> |

**CLO 3** Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing and Making Connections**  
**Oct 27 – 29** | 4.4 A Closer Look at Testing • 4.5 Making Connections | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Instructional Materials                                                                                                                                                                  | Learning Activities                                                                                                                                                                                    | Assessments                                                                                                                                  | Tools                                                                                            |
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| <p><b>1.</b> Describe Type I and Type II errors in the context of a study, relate <math>\alpha</math> to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CLO 4, CLO 8)</p> <p><b>2.</b> Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CLO 8)</p>                                                                                                                       | <p>e-Text: 4.4 A Closer Look at Testing</p> <p>Interactive Lecture: 4.4 A Closer Look at Testing</p> <p>Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)</p> | <p>Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions</p>                               | <p>Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM)</p>                                                           | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>                       |
| <p><b>3.</b> Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CLO 8)</p> <p><b>4.</b> Compare bootstrap and randomization distributions by their center and use, identifying each center for given data. (CLO 5)</p> <p><b>5.</b> Use a 95% confidence interval to reach a two-tailed test conclusion at <math>\alpha = 0.05</math>, and decide whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)</p> | <p>e-Text: 4.5 Making Connections</p> <p>Interactive Lecture: 4.5 Making Connections</p> <p>Slides: 4.5 Making Connections</p>                                                           | <p>Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples and where each distribution is centered</p> | <p>Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)</p>                                                                     | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>                       |
| <p><i>Items that cover objectives from more than one row above (objectives in parentheses)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                          |                                                                                                                                                                                                        | <p>Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4, 4.5 (all objectives)</p>                                                         | <p>Assignment Tool in Blackboard</p>                                                             |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |                                                                                                                                                                                                        | <p>Practice Test 2 (Blackboard, by Sun Nov 1)</p> <p>Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.</p> | <p>Assessment Tool in Blackboard</p> <p>Respondus</p> <p>LockDown</p> <p>Browser and Monitor</p> |

**CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 11: Normal Distributions and Inference for a Proportion**

**Nov 3 – 5** | 5.1 Hypothesis Testing Using Normal Distributions • 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Instructional Materials                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Learning Activities                                                                                                                                                                                                                                                                                            | Assessments                                                                                                                                                                        | Tools                                                                              |
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| <p><b>1.</b> Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution <math>N(\text{null value}, SE)</math>. (CLO 5)</p> <p><b>2.</b> Compute a standardized <math>z</math> statistic and find the <math>p</math>-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CLO 3, CLO 8)</p> <p><b>3.</b> Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CLO 5)</p> <p><b>4.</b> Construct and interpret a confidence interval for a population proportion with <math>z^*</math> for 90%, 95%, or 99% confidence. (CLO 6)</p> <p><b>5.</b> Carry out a one-proportion <math>z</math> test and conclude in context, and explain why the CI standard error uses <math>\hat{p}</math> while the test uses <math>p_0</math>. (CLO 8)</p> | <p>e-Text: 5.1 Hypothesis Tests Using Normal Distributions</p> <p>Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions</p> <p>Slides: 5.1 Hypothesis Tests Using Normal Distributions</p> <p>e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test)</p> <p>Interactive Lecture: 6.1 Inference for a Proportion</p> <p>Slides: 6.1 Inference for a Proportion</p> <p>StatKey – Theoretical Distributions (Normal)</p> | <p>Participation: Normal Distributions and <math>z</math> – malaria/mosquito example: compute <math>z</math>, shade the tail in StatKey, find the <math>p</math>-value</p> <p>Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs</p> | <p>Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM)</p> <p>Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>         |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                | <p>Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)</p>                                                                                                       | <p>Assessment Tool in Blackboard</p> <p>Respondus LockDown Browser and Monitor</p> |

**CLO 3** Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 12: Inference for Two Proportions and for a Mean**

**Nov 10 – 12** | 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) • 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                               | Instructional Materials                                                                                                                                                                                                                                      | Learning Activities                                                                                                                                                                  | Assessments                                                                               | Tools                                                   |
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| <b>1.</b> Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CLO 5)<br><b>2.</b> Construct and interpret a confidence interval for $p_1 - p_2$ , including what its sign means and whether it contains 0. (CLO 6)<br><b>3.</b> Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CLO 8) | e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT)<br>Interactive Lecture: 6.3 Inference for a Difference in Proportions<br>Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization) | Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula | Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM) | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom (live class) |
| <b>4.</b> Explain why inference for a mean uses the t distribution with $df = n - 1$ , and check the conditions for t procedures. (CLO 5)<br><b>5.</b> Construct and interpret a t confidence interval for a population mean, and carry out a one-sample t test. (CLO 6, CLO 8)                                                                                                                                 | e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT)<br>Interactive Lecture: 6.2 Inference for a Mean<br>Slides: 6.2 Inference for a Mean<br>StatKey – Theoretical Distributions (t)                                                                     | Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups                                                                                               | Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)                      | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom (live class) |
| <i>Items that cover objectives from more than one row above (objectives in parentheses)</i>                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                              |                                                                                                                                                                                      | Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 5.1, 6.1–6.3 (all objectives)        | Assignment Tool in Blackboard                           |

**CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 13: Inference for Two Means and Paired Data**

**Nov 17 – 19** | 6.4 Inference for a Difference in Means (Distribution, CI, Hypothesis Test) • 6.5 Paired Difference in Means | Fall 2026 · Section 036 · TTh  
6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                         | Instructional Materials                                                                                                                                                                    | Learning Activities                                                                                                              | Assessments                                                                                             | Tools                                                                   |
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| <b>1.</b> Compute the standard error and conservative degrees of freedom for a difference in two independent means, and check the conditions. (CLO 5)<br><b>2.</b> Construct and interpret a confidence interval for $\mu_1 - \mu_2$ , explaining how the order of the groups affects the sign. (CLO 6)<br><b>3.</b> Carry out a two-sample t test, choosing the tail from the research question. (CLO 8) | e-Text: 6.4 Inference for a Difference in Means (6.4-D, 6.4-CI, 6.4-HT)<br>Interactive Lecture: 6.4 Inference for a Difference in Means<br>Slides: 6.4 Inference for a Difference in Means | Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test                                               | Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)                     | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom (live class)                 |
| <b>4.</b> Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8)<br><b>5.</b> Define the difference variable d, carry out a paired t test and paired t confidence interval, and explain why pairing reduces variability. (CLO 6, CLO 8)                                                                                                     | e-Text: 6.5 Paired Difference in Means<br>Interactive Lecture: 6.5 Paired Difference in Means<br>Slides: 6.5 Paired Difference in Means                                                    | Participation: Paired or Independent? – class discussion of eight study scenarios (a)–(h); define d and the order of subtraction | Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)                              | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom (live class)                 |
| <i>Items that cover objectives from more than one row above (objectives in parentheses)</i>                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                            |                                                                                                                                  | Test 3 – Fri Nov 20 – Sun Nov 22 (Blackboard, proctored, 2 hours). Covers Weeks 10–13. (all objectives) | Assessment Tool in Blackboard<br>Respondus LockDown Browser and Monitor |

**CLO 1** Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 14: Chi-Square Test for Association**

**Nov 24 – 26** | 7.2 Testing for an Association Between Two Categorical Variables • Thanksgiving Holiday – No Class (Thu Nov 26) | Fall 2026 · Section 036  
· TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Instructional Materials                                                                                                                                                                                                                                                                                                       | Learning Activities                                                                                                                                                       | Assessments                                                                                                                                         | Tools                                                                                                                   |
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| <p><b>1.</b> State the hypotheses for a chi-square test of association from a two-way table. (CLO 8)</p> <p><b>2.</b> Compute expected counts and check that all expected counts are at least 5. (CLO 8)</p> <p><b>3.</b> Compute the chi-square statistic and <math>df = (r - 1)(c - 1)</math>, and interpret the right-tail p-value. (CLO 8)</p> <p><b>4.</b> Relate the chi-square test on a 2×2 table to the two-proportion z test, and explain that a significant result shows association but not direction or causation. (CLO 8)</p> | <p>e-Text: 7.2 Testing for an Association Between Two Categorical Variables</p> <p>Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables</p> <p>Slides: 7.2 Testing for an Association Between Two Categorical Variables</p> <p>StatKey – Theoretical Distributions (<math>\chi^2</math>)</p> | <p>Participation: Chi-Square Test – faculty rank × gender: compute expected counts, cell contributions, <math>\chi^2</math> and <math>df</math>; decide and interpret</p> | <p>Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)</p>                                 | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p>                                              |
| <p><i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                           | <p>Take-Home Quiz 8 (due Tue Nov 24, 11:59 PM) – covers 6.4, 6.5</p> <p>Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)</p> | <p>Assignment Tool in Blackboard</p> <p>Assessment Tool in Blackboard</p> <p>Respondus LockDown Browser and Monitor</p> |

**CLO 8** Perform hypothesis testing using statistical methods.

**MATH 1342 – Elementary Statistics – Week 15: ANOVA and Inference for Slope and Correlation**

**Dec 1 – 3** | 8.1 Analysis of Variance • 9.1 Inference for Slope and Correlation | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Instructional Materials                                                                                                                                                             | Learning Activities                                                                                                                                                                                                         | Assessments                                                                               | Tools                                                                      |
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| <p><b>1.</b> Identify when ANOVA is appropriate and state <math>H_0</math> and <math>H_a</math> in symbols and words. (CLO 8)</p> <p><b>2.</b> Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against <math>H_0</math>. (CLO 8)</p> <p><b>3.</b> Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CLO 8)</p>                                                                                                                                                                                                                                                                                                               | <p>e-Text: 8.1 Analysis of Variance</p> <p>Interactive Lecture: 8.1 Analysis of Variance</p> <p>Slides: 8.1 Analysis of Variance</p> <p>StatKey – Theoretical Distributions (F)</p> | <p>Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide</p>                                                                                                                       | <p>Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)</p>                | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p> |
| <p><b>4.</b> State the simple linear model, distinguish the population slope <math>\beta_1</math> and correlation <math>\rho</math> from the sample estimates <math>b_1</math> and <math>r</math>, and interpret the slope in context. (CLO 7)</p> <p><b>5.</b> Use regression output to test <math>H_0: \beta_1 = 0</math> (equivalently <math>\rho = 0</math>) with a t statistic on <math>n - 2</math> degrees of freedom, state the conclusion in context, and explain why the slope and correlation tests give the same p-value. (CLO 7, CLO 8)</p> <p><b>6.</b> Construct and interpret a confidence interval for the slope, interpret <math>R^2</math> as the proportion of variability explained, and check the conditions with a residual plot. (CLO 6, CLO 7)</p> | <p>e-Text: 9.1 Inference for Slope and Correlation</p> <p>Interactive Lecture: 9.1 Inference for Slope and Correlation</p> <p>Slides: 9.1 Inference for Slope and Correlation</p>   | <p>Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, <math>R^2</math>); self-quiz: drinks per week vs. classes missed</p> | <p>Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)</p> | <p>WileyPLUS</p> <p>StatKey</p> <p>Blackboard</p> <p>Zoom (live class)</p> |

**CLO 6** Describe and compute confidence intervals. **CLO 7** Solve linear regression and correlation problems. **CLO 8** Perform hypothesis testing using statistical methods.

# MATH 1342 – Elementary Statistics – Week 16: Putting It All Together and Final Exam

**Dec 8 – 10** | Review: Putting It All Together – From Sampling to Hypothesis Testing (Tue Dec 8) • Final Exam – Thu Dec 10, online (Blackboard, proctored)  
| Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Instructional Materials                                                                                                                                                | Learning Activities                                                                                                                                                      | Assessments                                                                            | Tools                                                                                                                                        |
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| <p><b>1.</b> Choose the correct procedure among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression, and carry out a complete six-step hypothesis test. (CLO 8)</p> <p><b>2.</b> Carry out and interpret a complete confidence interval or hypothesis test, using simulation or theoretical distributions as appropriate. (CLO 6, CLO 8)</p> <p><b>3.</b> Interpret results in context, including what the study design allows the conclusion to claim about generalization and causation. (CLO 1)</p> | <p>Review Slides: Putting It All Together (with answer key and no-answers versions)</p> <p>All Section Slides and Interactive Lectures (Weeks 1–15)</p> <p>StatKey</p> | <p>Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI</p> | <p>Final Exam – Thu Dec 10 (Blackboard, proctored, 2 hours, cumulative; no retake)</p> | <p>Blackboard</p> <p>StatKey</p> <p>Zoom (live class)</p> <p>Assessment Tool in Blackboard</p> <p>Respondus LockDown Browser and Monitor</p> |

**CLO 1** Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.